

ATIVIDADE

SEMÁFORO MILLIS()

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CÓDIGO FONTE:

```
#define TIME_GREEN 5000  
#define TIME_YELLOW 2000  
#define TIME_RED 1000
```

```
unsigned long time = 0;  
int position = 1;
```

```
bool pedidoPedestre = false;
```

```
void setup() {
```

```
    pinMode(2, INPUT_PULLUP);
```

```
    pinMode(3, OUTPUT);  
    pinMode(4, OUTPUT);  
    pinMode(5, OUTPUT);  
    pinMode(6, OUTPUT);  
    pinMode(7, OUTPUT);  
    pinMode(8, OUTPUT);  
    pinMode(9, OUTPUT);  
    pinMode(10, OUTPUT);  
    digitalWrite(3, HIGH);  
    digitalWrite(8, HIGH);  
    digitalWrite(9, HIGH);
```

```
digitalWrite(10, LOW);
```

```
time = millis();  
}
```

```
bool pressionado() {  
    return digitalRead(2) == LOW;  
}
```

```
void pedestre() {
```

```
    digitalWrite(5, HIGH);  
    digitalWrite(8, HIGH);
```

```
    digitalWrite(9, LOW);  
    digitalWrite(10, HIGH);
```

```
    delay(4000);
```

```
    digitalWrite(10, LOW);
```

```
    for (int i = 0; i < 4; i++) {  
        digitalWrite(9, HIGH);  
        delay(250);  
        digitalWrite(9, LOW);  
        delay(250);
```

```
}
```

```
digitalWrite(9, HIGH);
```

```
pedidoPedestre = false;
```

```
}
```

```
void loop() {
```

```
  if (pressionado()) {
```

```
    pedidoPedestre = true;
```

```
  }
```

```
  if (position == 1) {
```

```
    if (millis() - time >= TIME_GREEN) {
```

```
      position = 2;
```

```
      time = millis();
```

```
      digitalWrite(3, LOW);
```

```
      digitalWrite(4, HIGH);
```

```
    }
```

```
  }
```

```
  if (position == 2) {
```

```
if (millis() - time >= TIME_YELLOW) {  
  
    position = 3;  
    time = millis();  
  
    digitalWrite(4, LOW);  
    digitalWrite(5, HIGH);  
}  
}
```

```
if (position == 3) {  
  
    if (millis() - time >= TIME_RED) {  
  
        if (pedidoPedestre) {  
            pedestre();  
        }  
  
        digitalWrite(8, LOW);  
        digitalWrite(6, HIGH);  
  
        position = 4;  
        time = millis();  
    }  
}
```

```
if (position == 4) {
```

```
if (millis() - time >= TIME_GREEN) {
```

```
    position = 5;
```

```
    time = millis();
```

```
    digitalWrite(6, LOW);
```

```
    digitalWrite(7, HIGH);
```

```
}
```

```
}
```

```
if (position == 5) {
```

```
    if (millis() - time >= TIME_YELLOW) {
```

```
        position = 6;
```

```
        time = millis();
```

```
        digitalWrite(7, LOW);
```

```
        digitalWrite(8, HIGH);
```

```
    }
```

```
}
```

```
if (position == 6) {
```

```
    if (millis() - time >= TIME_RED) {
```

```
if (pedidoPedestre) {  
    pedestre();  
}
```

```
digitalWrite(5, LOW);  
digitalWrite(3, HIGH);
```

```
position = 1;  
time = millis();  
}  
}  
}
```

PRINT:

